

screen will open up, prompting you to enter all the details. Key in the details in the GUI panel that appears on the screen, as shown in Fig. 2.

Steps.

1. Press the different buttons as mentioned in the GUI. When the message button is pressed, input window will open, which means you need to enter a column size for random bit generation. Value of the column size can be any natural number of choice. By default, the value is ten.

2. After the message sequence is selected, message in the form of zeroes and ones is displayed on the screen. From the given code rates, you can select any code rate of your choice. This code rate will generate poly2trellis.

3. poly2trellis is used to accept the polynomial description of a convolutional encoder and return the corresponding trellis structure description. poly2trellis function includes constraint length and code generator.

Trellis = poly2trellis (constraint length, code generator)

4. Constraint length indicates the number of bits stored in the shift register, including the current bit. The code generator specifies the output connections for each encoder input bit stream.

Main parts of the GUI panel are shown in Fig. 2 and the description is given in the box.

5. The encoder will encode the message sequence and random noise will be added to the encoded message. At the decoder, the message is decoded using Viterbi algorithm.

6. At last, bit error is calculated, which indicates the number of errors in the decoded message sequence.

Steps for testing

1. Copy the complete folder (update) of this project from EFY DVD or the website to your system.

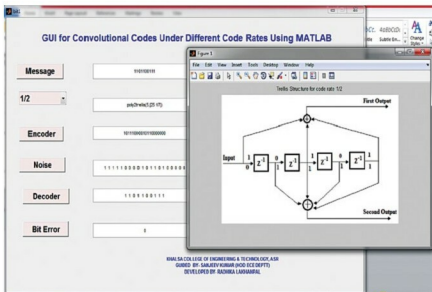


Fig. 3: GUI window with calculated bit error for code rate (1/2)

GUI Panel description

1. Click on the message button to select the desired column size.
2. Select the code rate of your choice from the given four code rates.
3. Trellis of corresponding code rate appears on the screen.
4. Message is encoded and random noise is added.
5. Click on Decoded button to decode the message.
6. Press Bit Error button to calculate the numbers of errors.
2. Run bit.m program.
3. The main GUI panel for CC opens up on the screen (Fig. 2).
4. Click on Message button. Input window will open, which asks you to enter the column size of any random message.
5. Select code rates out of the four defined. Select any one by pressing Code Rate button. Corresponding trellis design appears on the panel (Fig. 3).
6. Click on Encoder button to encode the message.
7. Click on Noise button to add random noise for inserting errors in the encoded message.
8. Here, the message is decoded by pressing Decoder button. Due to CC, errors inserted by noise are removed.
9. Finally, number of errors in the message is calculated by pressing Bit Error button.

Future application

CCs are used to combat transmission errors caused by a noisy channel. These allow communication to achieve transmission reliability by reducing bit error rate. It reduces the amount of power needed for transmission of data.

Viterbi algorithm allows safe data transmission via error correction. Original message can recover accurately without any noise. **EFY**

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

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